(although we can’t be sure, because we don’t have an answer to the *P vs NP* problem).

The first consequence for us of TSP being NP-hard is that we can’t assume we’ll be able to solve an instance of this problem on the fly, on a driver’s mobile phone or even a laptop computer (unless the number of cities to deliver to is small: but for the 10 cities in figure 17.8, there are already 10! ~ 3.6 million possible sequences) – we need computer power, and planning ahead, and possibly even computing ahead, and reusing the results as much as possible.

Even with a supercomputer at our service, in fact, the running time of an exact algorithm grows just too fast: with 15 cities there are ~1.3 trillion possible solutions, that becomes `2.4e18` (2 billion of billions) with 20 cities; even assuming that we can find an algorithm running in exponential time (which is asymptotically better than factorial), we couldn’t handle more than ~40 different deliveries (probably far less).

17.2.1 Exact vs Approximated Solutions

The last consideration we make is that we can afford to compromise on the optimal solution: we can be fine using a good near-optimal solution, we don’t need the best possible – for instance, if we add a few miles over a total of 1000 miles per trip, it feels acceptable, while doubling the total distance of the tour would be quite expensive, and making it 10 times costlier would be a disaster.